



PLATFORM9

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Summary

This doc elaborates the requirements and best practices for building an images specific to PCD.

General Best practices

Image formats

The preferred format for uploading images in glance is `QCOW2` format.

Uploading image with properties

- For any large images prefer uploading them from the `openstack` command line way:

```
openstack image create --os-interface admin --insecure --container-format bare --disk-format qcow2 --public
--file somdisk.img --property hw_disk_bus=scsi --property os_require_quiesce=yes --property
hw_qemu_guest_agent=yes --property hw_scsi_model=virtio-scsi --property hw_machine_type=q35 --property
os_type=linux some_file_name
```

Properties details:

- `hw_disk_bus=scsi` : This instruct the instance launch proces to pick up `scsi` for disk bus emulation
- `hw_scsi_model=virtio-scsi` : Needed for selecting the `scsi` model where `virtio-scsi` being more suitable
- `hw_qemu_guest_agent=yes` : Nova uses this to create `virtio-serial` device and channel to the instance. Needed by `qemu-guest-agent` service
- `os_require_quiesce=yes` : Nova that before taking a snapshot it must successfully call `guest-fsfreeze-freeze` to quiesce the application operations inside VM before a live-snapshot is performed
- `hw_machine_type=q35` : Fo selecting more modern hardware emulation during VM launch
- `os_type=linux` : should be changed to `os_type=windows` in case of windows OS in use. This is needed by nova to instruct `libvirt` service to precisely build the xml for the given OS

Package and service integration within images

Cloud-init services

Linux

- Most of the cloud images that are available from the official channel for respective OS like Ubuntu, RHEL, SLES will come with with cloud-init service pre-installed **which will be a required configuration to ensure the images can be customised as with external scripts and configurations at the time of launch.**

Windows

- The image needs to be pre-installed with `cloudbase-init` and `virtio` drivers. A detailed guide on building windows images with the required packages, to be consumed for deploying instance can be found in the official cloud director documentation: <https://docs.platform9.com/private-cloud-director/tutorials/create-windows-vm-from-iso/creating-a-windows-virtual-machine--vm--from-an-iso-image>

⚠ virtio-serial and the guest-agent service installation in Windows:

- Inside the mounted ISO navigate to `vioserial` directory select the windows version as needed
- Run the `vioser.inf` to install the serial drivers
- After successful installation of serial driers, navigate to `guest-agent` directory
- install the `qemu-ga-x86_64.msi`

In case of vjailbreak migrated VMs, these integration of drivers are taken care by the migration process when the VMs are moved and launched in PCD hypervisor.

Qemu-guest-agent services

This is a service when installed, will run inside the guest and communicates with the hypervisor host through a `virtio-serial` channel (out-of-band communication) which is used by nova service at the time of live-migration synchronisation, better guest details reporting mechanism to nova, avoid time drift within guest time sync etc.

The installation and enabling the service varies between different flavor of OS

Linux

RHEL/ROCKY

```
dnf install -y qemu-guest-agent
systemctl enable --now qemu-guest-agent
systemctl start qemu-guest-agent
```

SLES

```
zypper install -y qemu-guest-agent
systemctl enable --now qemu-guest-agent
systemctl start qemu-guest-agent
```

Ubuntu

```
apt install -y qemu-guest-agent
systemctl enable --now qemu-guest-agent
systemctl start qemu-guest-agent
```

Windows

The Installation process is elaborated in the [cloud-init install section](#)

Network service within instance images

In general the networking for the instance level OS is expected to be set as DHCP which will allow the VM to pick the IP that is provided by the underlying neutron service via DHCP/BOOTP ip assignment (with or without port creation operation on the neutron level)

To help support the default way of IP assignment, the image can be built with a generic netplan like below which will ensure the IP gets assigned once the `firstboot` script is triggered when the VM is launched:

```
# /etc/cloud/cloud.cfg.d/50-network-config.cfg

network:
  version: 2
  ethernets:
    default-nic1:
      match:
        name: "en*"           # Matches first available en* interface
      dhcp4: true
      dhcp4-overrides:
        use-hostname: false
        use-dns: true
        use-routes: true
```

```
    route-metric: 100
critical: true
optional: true
nic2:
  match:
    name: "en*"          # cloud-init assigns remaining interfaces
  dhcp4: true
  dhcp4-overrides:
    use-hostname: false
    use-dns: false
    use-routes: false
    route-metric: 200
critical: false
optional: true
nic3:
  match:
    name: "en*"
  dhcp4: true
  dhcp4-overrides:
    use-hostname: false
    use-dns: false
    use-routes: false
    route-metric: 300
critical: false
optional: true
```